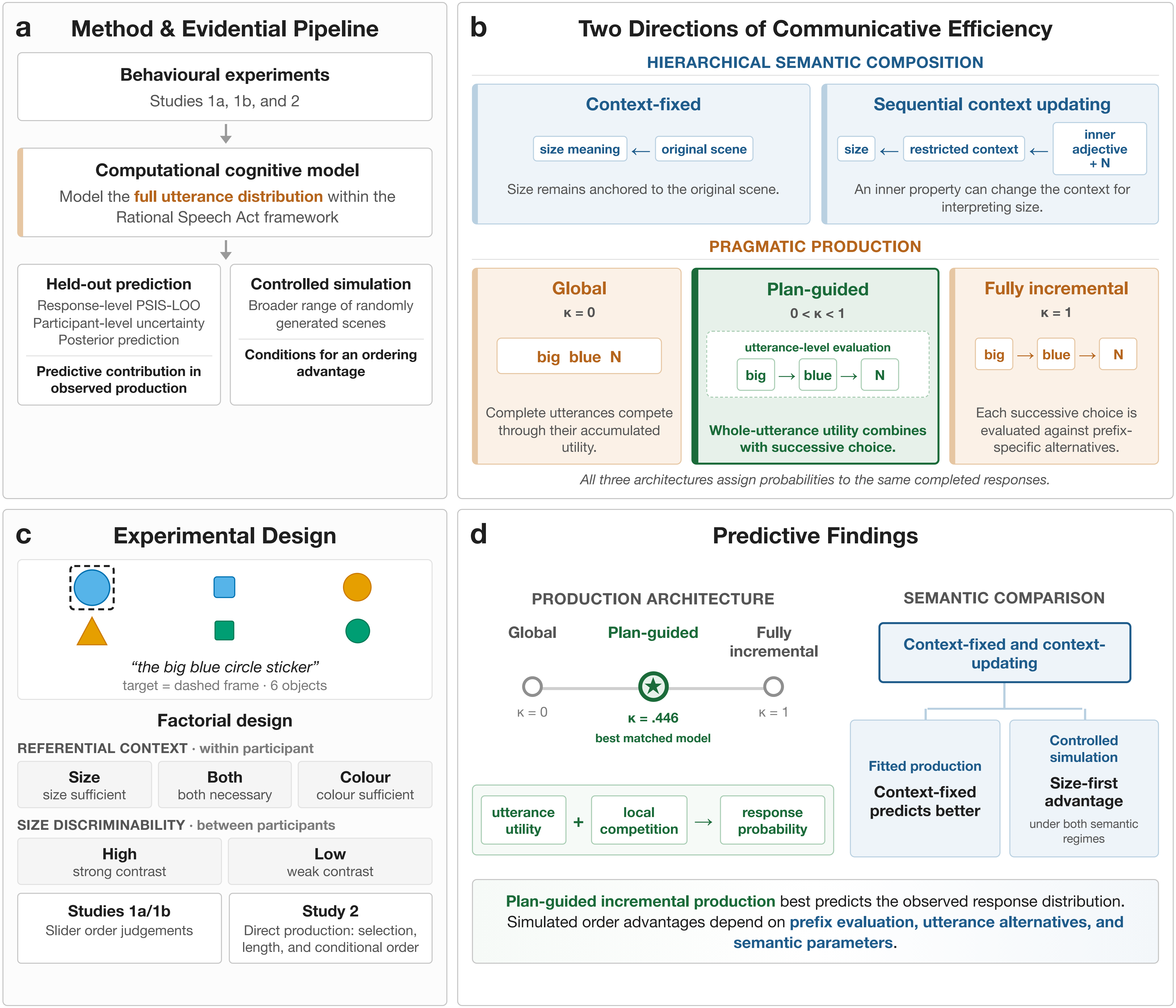
a
Method & Evidential Pipeline
Behavioural experiments
Studies 1a, 1b, and 2
Computational cognitive model
Model the full utterance distribution within the Rational Speech Act framework
Held-out prediction
Response-level PSIS-LOO
Participant-level uncertainty
Posterior prediction
Predictive contribution in observed production
Controlled simulation
Broader range of randomly generated scenes
Conditions for an ordering advantage
b
Two Directions of Communicative Efficiency
HIERARCHICAL SEMANTIC COMPOSITION
Context-fixed
size meaning ⟵ original scene
Size remains anchored to the original scene.
Sequential context updating
size ⟵ restricted context ⟵ inner adjective + N
An inner property can change the context for interpreting size.
PRAGMATIC PRODUCTION
Global
κ = 0
big blue N
Complete utterances compete through their accumulated utility.
Plan-guided
0 < κ < 1
utterance-level evaluation
big ⟶ blue ⟶ N
Whole-utterance utility combines with successive choice.
Fully incremental
κ = 1
big ⟶ blue ⟶ N
Each successive choice is evaluated against prefix-specific alternatives.
All three architectures assign probabilities to the same completed responses.
c
Experimental Design
“the big blue circle sticker”
target = dashed frame · 6 objects
Factorial design
REFERENTIAL CONTEXT · within participant
Size
size sufficient
Both
both necessary
Colour
colour sufficient
SIZE DISCRIMINABILITY · between participants
High
strong contrast
Low
weak contrast
Studies 1a/1b
Slider order judgements
Study 2
Direct production: selection, length, and conditional order
d
Predictive Findings
PRODUCTION ARCHITECTURE
Global
Plan-guided
Fully incremental
κ = 0
κ = .446
best matched model
κ = 1
utterance utility + local competition ⟶ response probability
SEMANTIC COMPARISON
Context-fixed and context-updating
Fitted production
Context-fixed predicts better
Controlled simulation
Size-first advantage
under both semantic regimes
Plan-guided incremental production best predicts the observed response distribution. Simulated order advantages depend on prefix evaluation, utterance alternatives, and semantic parameters.